# Towards Assessing the Impact of TWENDE Project on Climate Resilience, Incomes and Livestock Units of Pastoral Communities in Kenya: Baseline estimates using Propensity Score Matching and Difference-in-Differences Analyses.

**Francis M. Musau**

International Union for Conservation of Nature (IUCN)

Eastern and Southern Africa Regional Office (ESARO) Nairobi, Kenya.

Francis.Musau@iucn.org

Towards Assessing the Impact of TWENDE[1] Project on Climate Resilience, Incomes and Livestock Units of Pastoral Communities in Kenya: Baseline estimates using Propensity Score Matching and Difference-in-Differences Analyses.

**Francis M. Musau**
IUCN ESARO Nairobi, Kenya.
Francis.Musau@iucn.org

**Abstract:** Assessment of the impact of a climate adaptation project (TWENDE) on resilience, incomes and livestock units of pastoral communities used quasi-experimental methods. Specifically, the study employed propensity score matching (PSM) and difference-in-differences (DiD) analyses in Stata 17 software. We used PSM to address the limitation of selection bias by creating an artificial comparison group that has similar characteristics to the treatment group. DiD estimated the project impact by computing the difference in project outcomes before and after the project and between the treatment and control groups. At baseline, the results showed that males were more likely to participate in the project while people that were more elderly, more educated and married were less likely to participate. The study also found that the mean household resilience index, monthly incomes and tropical livestock units of the treatment group were lower than those of the control group were. We concluded that (i) the project is intervening in the more vulnerable communities; (ii) the project design was valid with regards to the planned interventions; (iii) the identified comparison group is a valid control group that is suitable for estimation of the impact of the intervention at the end of the project; and (iv) households main occupation influences the choice of project interventions in which to enrol or participate.

**Key words:** *Difference-in-Differences, Impact, Kenya, Propensity score matching, TWENDE.*

## 1. Introduction

Climate change is as an existential risk, which threatens both human lives and future development (Spratt & Dunlop, 2019). The Intergovernmental Panel on Climate Change (IPCC, 2021) projects that global warming will be as high as 1.5°C by the year 2040 with adverse impacts such as the collapse of life-supporting ecosystems and consequently, the loss of related ecosystem services (e.g. water, food etc.) and economies and increased risk of climate disasters such as drought, floods and cyclones. Largely, ecosystems services underpin Africa's economy. For instance, climate change could potentially endanger 25 to 40% of mammal species in national parks (IPCC, 2007) thus affecting adversely tourism-dependent revenue and livelihoods. Climate-affected affected communities to lack the requisite productive and livelihoods assets thus becoming poorer and more vulnerable.

In Kenya, the collective economic cost of a drought and flood events is estimated at approximately US$4 billion and the overall economic cost of climate change is equivalent to a

[1] *TWENDE: Towards Ending Drought Emergencies.*

loss of almost 3% of GDP each year by 2030 (SEI, 2009) while the annual costs of climate adaptation is estimated at around $23 million/year by 2050 (Watkiss, Downing, & Dyszynski, 2010). It is against this background that the TWENDE project was designed specifically to reduce the cost of climate change induced drought on Kenya's national economy by increasing resilience of the livestock and other land use sectors in restored and effectively governed rangeland ecosystems. The project will achieve this through implantation of climate adaption and resilience interventions in three priority landscapes spread across 11 counties. These interventions include awareness and capacity building in climate change impacts and response measures; Sustainable Land Management practices (e.g. tree planting/reforestation, soils and water conservation, seed banking etc.); restoration; enterprises and/or climate-resilience value chains; planned; planned grazing and access to dry grazing/watering sites.

Building climate resilience requires intervention that simultaneously strengthen the absorptive (coping), adaptive (adjust or response), and transformative (system-level change) capacities at household, community or landscape levels (Asmamaw, Mereta, & Ambelu, 2019). The above-mentioned project interventions contribute directly to these three capacities.

## 2. Impact Theory of Change

Figure 1 below shows a simplified illustration of the roadmap through which change will happen as a result of TWENDE project interventions. The impact causal pathway of the project starts with the implementation of key project interventions (activities) which upon completion, will produce the project outputs of capacities (knowledge and skills), tools and infrastructure (grey and green) for building climate resilience. In the medium term, proper use/practise of the capacities and tools, coupled with equitable access and utilization of the productive assets/infrastructure, the outputs will result in the expected project outcomes of increased household assets, incomes and ultimately, increased resilience in the long term.

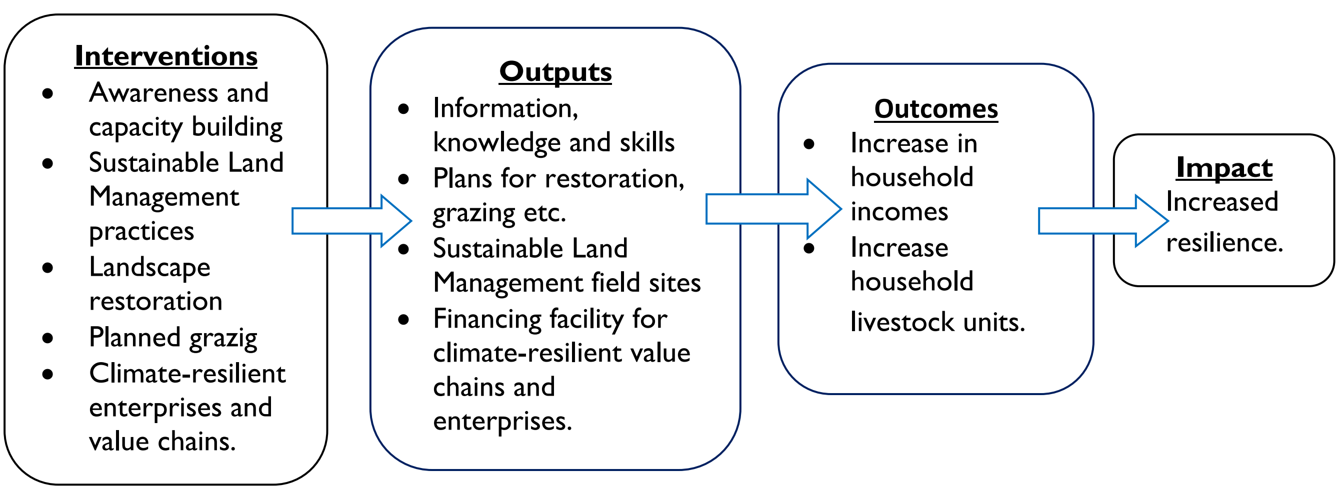


Figure 1: Intervention logic causal pathway.

However, the application of acquired information, knowledge and skills on the one hand, and the access and proper utilization, operation and maintenance of productive assets on the other, will depend not only on characteristics observed on the data of the target beneficiaries but also on unobserved variables. Other than the outcome/impact household characteristics that are affected by the project interventions, other observables that affect the knowledge and practice of climate resilience tools and approaches include household head health status (Asmamaw, Mereta, & Ambelu, 2019), gender,  (Dassanayake, Mohapatra, K, & Adamowicz, 2017), age, education and marital status of household heads. The unobserved variables that could potentially affect project outcomes are motivation, perception ability and management skills in climate resilience (Kiani, et al., 2021). To ensure there is no selection bias between the control and treatment groups, the study will control for the observable covariates.

## 3. Material and Methods

### 3.1. Study Area

This baseline study covered the three Twende project priority landscapes; Sabarwawa, Mid Tana and Kyulu (1b, 1a and 2 respectively in figure 2). These landscapes represent the two major climate zones of ASALS, the arid (Sabarwawa and Mid Tana) and semi-arid lands (Kyulu) which are characterized by extremely high inter-annual variability of rainfall. In Sabarwawa landscape, the communities are predominantly pastoralist while in both Mid Tana and Kyulu, they mixed pastoralists and agro-pastoralists.

Sabarwawa landscape covers three counties namely Samburu, Isiolo and Marsabit while Mid Tana is a transition between the arid lowlands and the humid highlands of Mount Kenya covering five counties (Wajir, Garissa, Tana River, Meru, Tharaka Nithi and Kitui). Both Sabarwawa and Mid Tana provide seasonal grazing for the neighbouring counties and for counties further to the north as far as in southern Ethiopia. Kyulu landscape spreads across the three counties of Taita Taveta, Kajiado and Makueni. It is part of the wider Amboseli ecosystem, which includes the Kyulu National Park and Kyulu Forest Reserve.

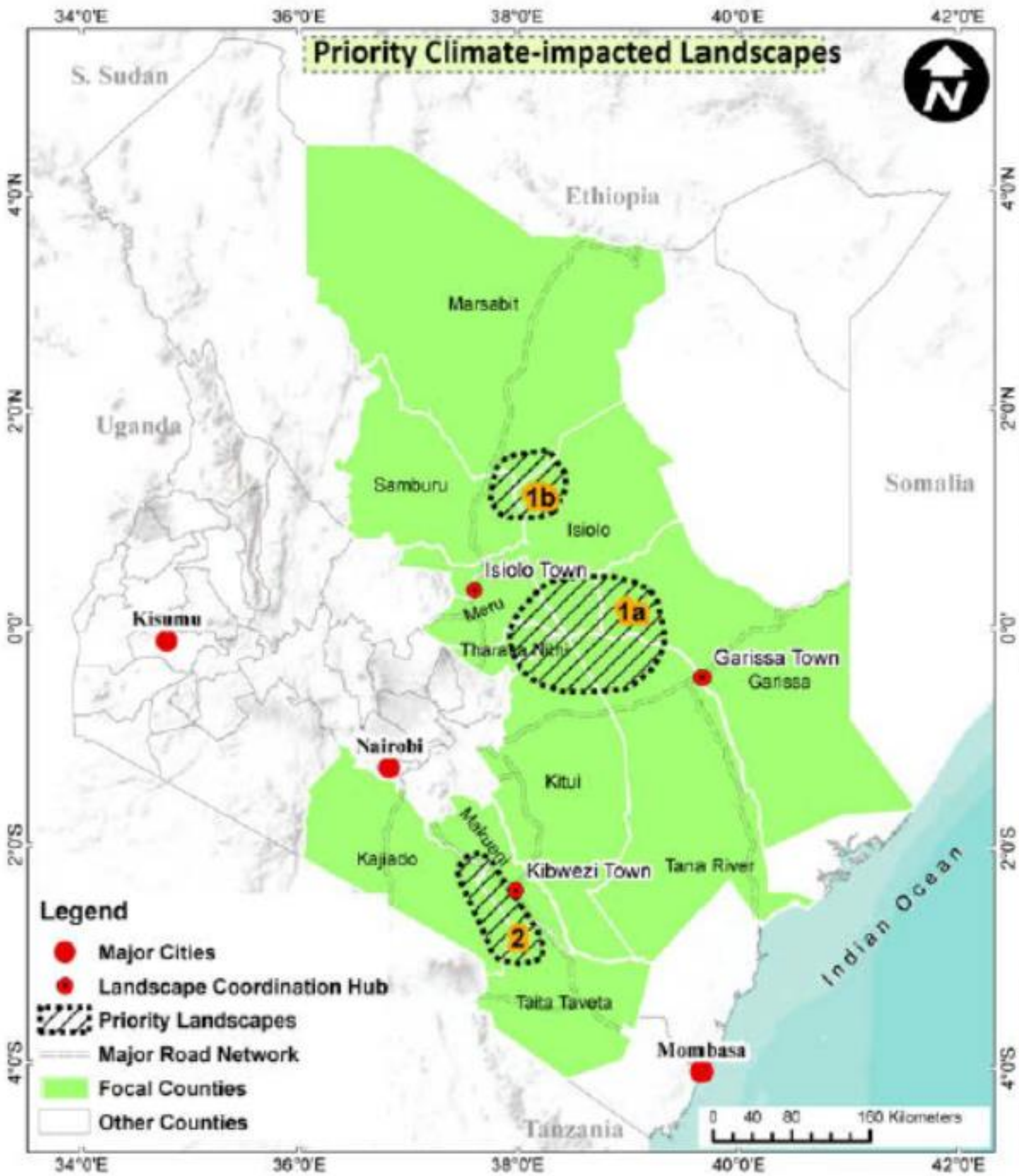


Figure 2: TWENDE project target landscapes, (Source: **(IUCN, TWENDE Towards Ending Drought Emergencies: Ecosystem Based Adaptation in Kenya's Arid and Semi-Arid Rangelands, 2015)**.

### 3.2 Sampling Approach

The data presented in this paper is the first (baseline) of two datasets for evaluating the impact of TWENDE project on household livestock assets, incomes and climate resilience. The second dataset (i.e. end line), will be collected at the end project implementation. The baseline data was collected between July and September 2021.

The sample size for the household survey was determined using power analysis in G*Power version 3.1.9.4 **(Faul & Edgar, 2007)**. The parameters used to compute the sample size for a two-tailed test (i.e. difference between two means) were effect size, Type I error probability, power and treatment/control group ratio (figure 3).

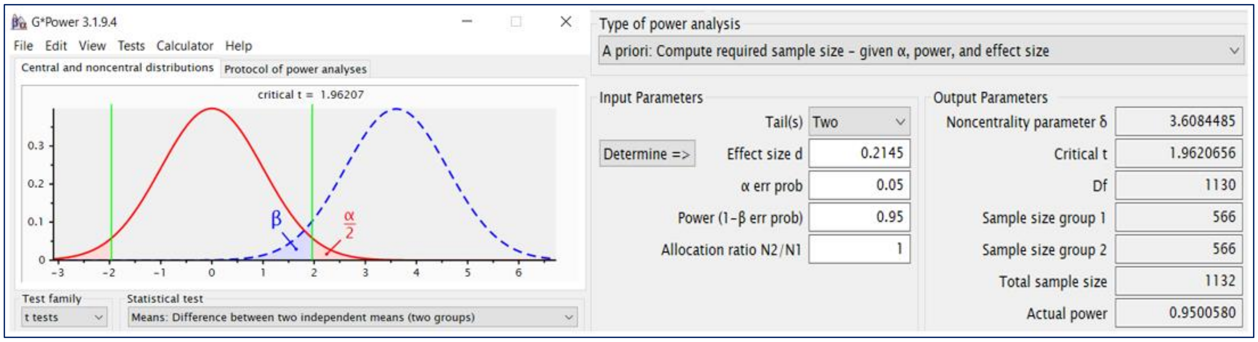


Figure 3: G*Power sample size estimation parameters.

This test generated 566 respondents for each of the two groups (treatment and control), producing an overall sample size of 1,132 respondents. To provide for non-responses during data collection or missing data, we added 10% making the total sample size to be 1,246, which was shared equally between the treatment and control groups (i.e. 623 household each). In addition to this, we also increased the samples size by an additional 46 households in order to ensure that each of the 11 counties had a sample size of more than 30 households for each group. This increased the total sample size to 1,292 respondents.

We used a two-stage sampling approach to identify the study respondents. In the first stage was the selection of two administrative wards in each county within the three landscapes, one of the wards being the "treatment" and the other "control". The wards were selected based on known or perceived similarity in socio-economic, ecological and climatic conditions and vulnerability to climate disasters such as drought and floods. The second stage involved random selection of respondent households from both treatment and control groups.

### 3.3 Evaluation Design

Randomization of the selection of TWENDE project beneficiaries (treatment groups) was not feasible because of spill over effects within the landscapes. Furthermore, considering that climate change affects all communities in the target landscapes, randomization would trigger political or ethical concerns. On this basis, we did not use Randomized Controlled Trials (RCT) design but instead, adopted a quasi-experimental design. However, this design is prone to suffer selection bias because the households or individuals participating in the project usually differ in observable and unobservable ways **(White & Raitzer, 2017)** hence making the estimate of the counterfactual inaccurate. To address this bias, we used the he propensity score matching (PSM) approach, which mimics randomization of treatment and control groups by choosing a comparison group that have similar probabilities (i.e. propensity score) to participate or enrol in the project **(Gertler, Martinez, Premand, L.B., & Vermeersc, 2016)**. PSM helps clone a valid counterfactual by creating a treatment and control groups that are similar observable characteristics.

The steps in this approach include defining the eligible population; identifying the treatment group(s) (based on eligibility or beneficiary selection criteria); selection of the control/comparison group (i.e. a group that is similar in all observable aspects to the treatment group except that the it doesn't participate in the project); using a probit model, estimate the probability (propensity score) that each individual in treatment and control groups participates in the project; and match groups/individuals based on defined common support (overlap) of propensity scores for the two groups (figure 4). After matching, the sample size reduced from 1292 to 1284 households.

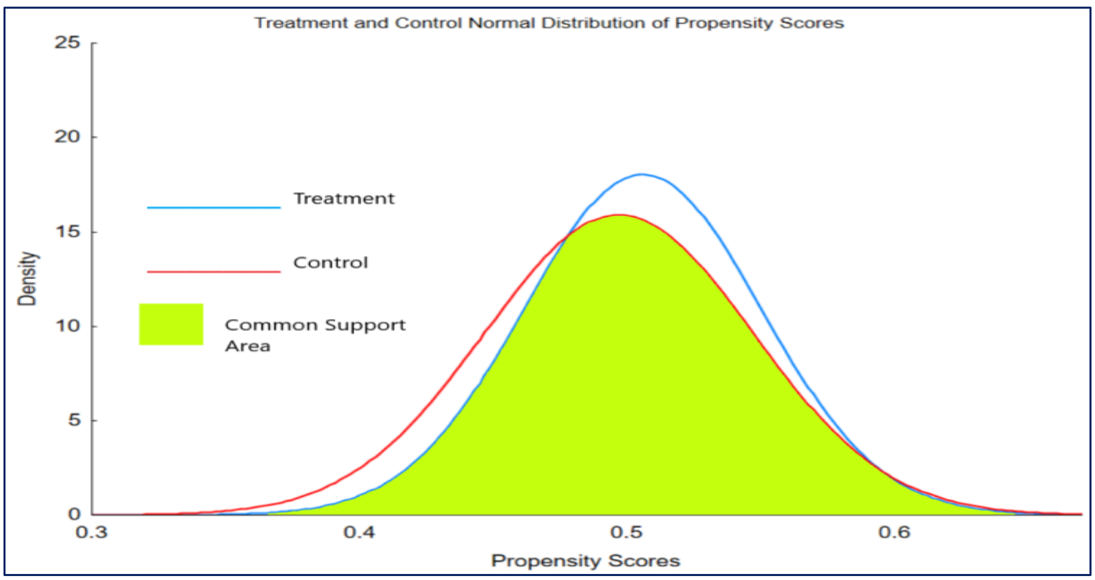


Figure 4: Propensity score Matching estimation of common support area.

Combining propensity score matching with other methods produces more robust results **(Gertler, Martinez, Premand, L.B., & Vermeersc, 2016)** and removes further the time-invariant unobserved characteristics that correlate with project outcomes **(Kiiru, 2019)**. For many years, impact evaluation experts have combined propensity score matching with Difference-in-Differences (DiD). The impact of a rural road program on community-level local market development in Vietnam was estimated using PSM with DiD **(Mu & Van de Walle, 2011)**; more recently, the Impact of project interventions on post-harvest loses among maize farmers in Tanzania was assessed also using Matched DiD **(Kiiru, 2019)**; and, an evaluation of the impact of climate resilience program on production and quality of life for smallholder farmers in Brazil was conducted by **(Gori Maia, Burney, Morales Martı´nez, & Cesano, 2021)**. Matching combined with difference-in-differences addresses the unobserved characteristics in the treatment and control groups **(Gertler, Martinez, Premand, L.B., & Vermeersc, 2016)**. If the "parallel trends" assumption is true that the outcome variable follows the same trajectory over time in both the treatment and comparison groups without the intervention, then DiD provides an unbiased estimate of the impact of the project **(White & Raitzer, 2017)**.

## 3.4 Data Collection on Outcome Indicators

### *3.4.1 Household Survey*

We conducted a household survey covering 1292 households in the three landscapes. The household survey questionnaire captured data on access to water, number of livestock, access to dry season grazing and watering areas, household monthly income and social capital i.e. household membership in women groups, youth groups, water resource users, community forest associations, merry-go-round, trader associations, savings and loan associations, grazing committees, rangeland management committees, disaster risk reduction committees and cooperative or commodity/producer marketing groups.

### *3.4.2 Climate Resilience Index (CRI)*

Household climate resilience index was computed from six variables namely (i) months accessing water from a reliable source; (ii) monthly household income (USD); (iii) months accessing dry season grazing/watering areas; (iv) household Tropical Livestock Units (TLUs); (v) months it takes to restock after drought; and (vi) number of groups or organizations in which the household is a member. Each of these variables was scored on a relative scale of 1 to 3 (1=Low, 2=Moderate, 3=High). For each household, the scores of the six variables were summed up and divided by six to obtain the mean composite climate resilience index.

### *3.4.3 Tropical Livestock Units (TLUs)*

The livestock data collected from household survey was converted into Tropical Livestock Units (McPeak & Little, 2017) where one (1) TLU represents 250 kg live weight of an animalis and is equivalent to 0.7 camel, 1 head of cattle, 10 sheep or 10 goats (Schwartz, Shaabani, & and Walther, 1991). The computation of TLUs did not include donkeys, poultry and beehives.

## 4.0 Results

## 4.1 Description of Respondents

Respondents were aged between 18-35 years (45%) and 36-60 years (43%). Only 11% were more than 60 years old. Education levels were low with more than 70% of the respondents having only primary level education or no education at all. The main occupations of the household head included livestock keepers (36.5%), crop farmers (30.9%), entrepreneurs

(12.5%), casual labour (11.6%) and wage employment (4.8%). More than half of the respondents (54%) were females, 77% of them married.

All households were members in at least one social organization, the main organizations in which households were members being merry-go-rounds (55%), women groups (49%), savings and loan associations (29%), youth groups (21%), peace committees (18%), water resource users (14%), commodity or producer marketing/trader associations (14%) and grazing committees (8%).

### 4.2 Propensity Score Matching (PSM)

For each of the three outcome indicators (Climate Resilience Index, Household Monthly Incomes and Household Livestock Units), PSM established a satisfactory test of the *Balancing Property of Propensity Scores* and that the mean scores for treatment and control groups were not different (p=0.986). At baseline, the results showed that males were more likely to participate in the project while people that were more elderly, more educated and married were less likely to participate.

Table 1: Estimation of Propensity Scores controlled for Age, Sex, Education and Marital Status.

| Probit regression | | | | Number of obs | = | 1292 |
|---|---|---|---|---|---|---|
| | | | | LR chi2(4) | = | 11.65 |
| | | | | Prob > chi2 | = | 0.0202 |
| Log likelihood = -889.71527 | | | | Pseudo R2 | = | 0.0065 |

| Outcomes | Coefficient | Std. err. | z | P>\|z\| | [95% conf. | interval] |
|---|---|---|---|---|---|---|
| Age | -.0403329 | .0535743 | -0.75 | 0.452 | -.1453367 | .0646708 |
| Sex | .057876 | .0723457 | 0.80 | 0.424 | -.083919 | .1996709 |
| Educ | -.1090714 | .0400256 | -2.73 | 0.006 | -.1875201 | -.0306227 |
| Marr | -.1209188 | .0581155 | -2.08 | 0.037 | -.2348232 | -.0070144 |
| _cons | .3829524 | .220716 | 1.74 | 0.083 | -.049643 | .8155477 |

### 4.3 Baseline Values of the Outcome Indicators.

#### *4.3.1 Climate Resilience Index*

Overall, the climate resilience index was low, with a mean score of 1.28. It was slightly higher in the control group than in the treatment group. The baseline difference-in-difference was negative 0.061implying that the average climate resilience index of the treatment group was less than that of the control group by 0.061 (table 2). The DiD estimates for the landscapes were 0.026, -0.089 and -0.122 for Kyulu, Mid Tana and Sabarwawa respectively.

Table 2: Difference-in-Difference estimates of climate resilience.

DIFFERENCE-IN-DIFFERENCES ESTIMATION RESULTS

Number of observations in the DIFF-IN-DIFF: **1284**

| | Before | After | |
|---|---|---|---|
| Control: | **636** | **0** | 636 |
| Treated: | **648** | **0** | 648 |
| | 1284 | 0 | |

| Outcome var. | CRI | S. Err. | \|t\| | P>\|t\| |
|---|---|---|---|---|
| Before | | | | |
| Control | **1.312** | | | |
| Treated | **1.251** | | | |
| Diff (T-C) | **-0.061** | **0.019** | **-3.23** | **0.001*** ** |
| After | | | | |
| Control | **1.312** | | | |
| Treated | **1.251** | | | |
| Diff (T-C) | **-0.061** | **0.019** | **3.23** | **0.001*** ** |
| Diff-in-Diff | **0.000** | . | . | . |

R-square: **0.01**

* Means and Standard Errors are estimated by linear regression

**Inference: *** p<0.01; ** p<0.05; * p<0.1

Although the climate resilience index for treatment and control groups were significantly different ($p<0.01$), both indices were generally low on a scale of 1 to 3 (1=Low, 2=Moderate, 3=High). Comparing the resilience index in the three landscapes, Sabarwawa had the highest while Mid Tana had the lowest (figure 5).

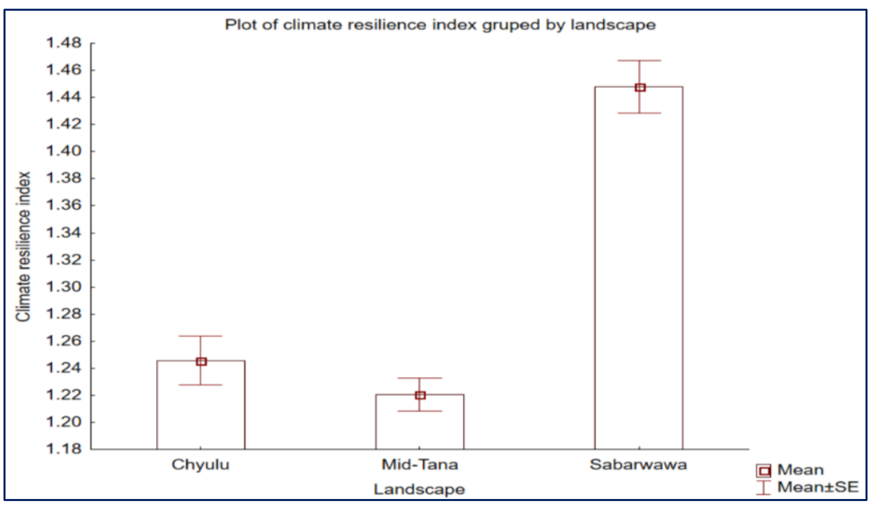


Figure 5: Climate resilience index in target landscapes.

The variation in the resilience indices among the three landscapes was statistically significant (df=2, F=50.96, $p<0.001$). Performing a post-hoc test revealed that the resilience index of Sabarwawa landscape was significantly higher than that of the other two landscapes although the resilience estimates of Kyulu and Mid Tana were not significantly difference from each other. The resilience index varied significantly among the counties with Taita Taveta, Makueni and Garissa being the least climate resilient while Kajiado, Samburu, Isiolo and Samburu had the highest resilience scores (table 3). In both Mid Tana and Sabarwawa, the control group had a higher resilience index than the treatment group. However, in Kyulu, the treatment group had a higher resilience.

Table 3: County climate resilience index.

| County | Taita Taveta | Makueni | Garissa | Kitui | Tharaka Nithi | Meru | Tana River | Marsabit | Samburu | Isiolo | Kajiado | All Groups |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Mean | 1.04 | 1.07 | 1.08 | 1.16 | 1.21 | 1.25 | 1.34 | 1.37 | 1.42 | 1.44 | 1.48 | 1.28 |
| N | 61 | 167 | 60 | 180 | 113 | 60 | 60 | 84 | 153 | 162 | 184 | 1284 |
| Std. Dev | 0.234 | 0.181 | 0.228 | 0.259 | 0.302 | 0.241 | 0.269 | 0.279 | 0.332 | 0.380 | 0.391 | 0.339 |

### *4.3.2 Household Monthly Incomes (US$)*

In the entire study area the average household income was estimated at US$94.73. Control groups had higher monthly income compared to the treatment group. Baseline DiD was US$-58.332 (table 4) indicating that the income level of treatment group was approximately USD58 lower than that of the control group.

Table 4: Difference-in-Difference estimates of household incomes.

```
DIFFERENCE-IN-DIFFERENCES ESTIMATION RESULTS
Number of observations in the DIFF-IN-DIFF: 1284
            Before          After
   Control: 636             0            636
   Treated: 648             0            648
            1284            0
---------------------------------------------------------
 Outcome var.  | Incomes | S. Err. |   |t|   |  P>|t|
---------------------------------------------------------
Before         |         |         |         |
   Control     | 124.173 |         |         |
   Treated     | 65.841  |         |         |
   Diff (T-C)  | -58.332 | 9.739   | -5.99   | 0.000***
After          |         |         |         |
   Control     | 124.173 |         |         |
   Treated     | 65.841  |         |         |
   Diff (T-C)  | -58.332 | 9.739   | 5.99    | 0.000***
               |         |         |         |
Diff-in-Diff   | 0.000   |    .    |    .    |    .
---------------------------------------------------------
R-square:    0.03
* Means and Standard Errors are estimated by linear regression
**Inference: *** p<0.01; ** p<0.05; * p<0.1
```

Similarly, control groups had higher incomes than treatment groups in the three landscapes. The highest income was recorded in Sabarwawa (US$108.58) followed by Kyulu (US$104.62) and Mid Tana (US$80.50). The pre-intervention DiD values were US$-83.01 in Kyulu, US$-63.07 in Mid Tana and US$-8.099 in Sabarwawa. These baseline income differences were statistically significant in both Kyulu and Mid Tana landscapes ($p<0.001$).

Income distribution in the counties also showed a significant variation where Taita Taveta, Tana River and Samburi had the lowest incomes while Kitui, Isiolo and Kajado had the highest (table 5).

Table 5: County household monthly income.

| County | Taita Taveta | Tana River | Samburu | Tharaka Nithi | Marsabit | Meru | Garissa | Makueni | Kitui | Isiolo | Kajiado | All Groups |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Mean | 23.10 | 36.67 | 58.65 | 61.00 | 64.94 | 65.23 | 86.43 | 90.01 | 128.00 | 141.20 | 144.90 | 94.73 |
| N | 61 | 60 | 153 | 113 | 84 | 60 | 60 | 167 | 180 | 162 | 184 | 1284 |
| Std. Dev | 17.09 | 22.83 | 69.67 | 84.69 | 39.66 | 84.27 | 105.33 | 113.02 | 348.78 | 188.52 | 176.80 | 176.83 |

### *4.3.3 Household Tropical Livestock Units.*

The average household livestock units were 16.41, the value being higher in the comparison group than the treatment group (table 6). The baseline DiD between the treatment and control group -1.804, which was not statistically significant (p>0.1).

Table 6: Difference-in-Difference estimates of household tropical livestock units.

```
DIFFERENCE-IN-DIFFERENCES ESTIMATION RESULTS
Number of observations in the DIFF-IN-DIFF: 1284
            Before         After
   Control: 636            0            636
   Treated: 648            0            648
            1284           0
--------------------------------------------------------
 Outcome var.     | TRUs     | S. Err. |   |t|   |  P>|t|
--------------------------------------------------------
Before            |          |         |         |
   Control        | 17.318   |         |         |
   Treated        | 15.514   |         |         |
   Diff (T-C)     | -1.804   | 3.434   | -0.53   | 0.600
After             |          |         |         |
   Control        | 17.318   |         |         |
   Treated        | 15.514   |         |         |
   Diff (T-C)     | -1.804   | 3.434   | 0.53    | 0.600
                  |          |         |         |
Diff-in-Diff      | 0.000    |    .    |    .    |    .
--------------------------------------------------------
R-square:    0.00
* Means and Standard Errors are estimated by linear regression
**Inference: *** p<0.01; ** p<0.05; * p<0.1
```

It was found that Mid Tana and Sabarwawa had higher TLU (21.91 and 16.61 respectively) while Kyulu had the least (8.58). In both Kyulu and Mid Tana, the control group had higher TLUs although the differences were not of statistical significance.

Variations in tropical livestock units were also observed in the counties where Tana River, Garissa and Isiolo recorded higher TLUs while relatively lower ones occurred in Taita Taveta, Tharaka Nithi and makueni (table 7).

Table 7: County household tropical livestock units.

| County | Taita Taveta | Tharaka Nithi | Makueni | Kitui | Samburu | Marsabit | Kajiado | Meru | Tana River | Garissa | Isiolo |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Mean | 1.93 | 3.25 | 3.72 | 3.99 | 8.35 | 8.71 | 15.20 | 27.20 | 37.83 | 41.65 | 49.42 |
| N | 61 | 113 | 167 | 180 | 153 | 84 | 184 | 60 | 60 | 60 | 162 |
| Std. Dev | 3.75 | 4.53 | 7.79 | 4.88 | 11.67 | 13.42 | 19.80 | 98.59 | 50.42 | 61.31 | 114.61 |

## 4.4 Pre-treatment Effects of Planned Interventions on the Project Outcomes.

### *4.4.1 Planned Grazing.*

Pre-project intervention, 25.5% of the households in the three landscapes reported that they were already practicing planned grazing. Comparing the households with those that did not practice this initiative, we found that households that practised planned grazing had climate resilience higher by 0.21 ($p<0.001$), monthly incomes higher by US$38.24 ($p<0.001$) and livestock numbers higher by approximately five units ($p>0.01$). Disaggregating the findings by treatment and control areas, the proportion of households engaging in planned grazing were 21.5% and 29.6% respectively. In both groups, households practising this intervention had higher resilience index and higher incomes. The association between the intervention and livestock units was also positive in the comparison group, thus indicating a positive association between planned grazing and the three outcomes.

### *4.4.2 Tree Planting for Restoration of Degraded Lands.*

In the three target landscapes, 34.9% of the households practice tree planting as a means to restore degraded landscapes. This represents 30.6% and 39.3% in the treatment and control areas respectively. The intervention was positively associated with both climate resilience and household incomes although the association was not statistically significant. Regarding the effect of tree planting on TLUs, we established that the intervention was associated with lower number of livestock units in both treatment and control groups. Overall, households practising tree planting had their livestock numbers lower by seven units ($p<0.001$), indicating either that household that enrol in tree planting interventions have lower number of livestock or that that tree planting may not be compatible with livestock production. Majority (55.4%) of the households that planted trees were crop farmers while only a few (19.2%) livestock keepers practised the intervention.

### *4.4.3 Bee Keeping.*

The percentage of households practising bee keeping is 12.0% in the general target area, 11.4% in treatment and 12.6% in control areas. The intervention is positively associated with both climate resilience and household incomes. However, a negative association was found between bee keeping and household livestock units ($p<0.05$). Overall, households that engage in apiculture have livestock assets that are less by approximately six units. This could possibly be an indication that both bee keeping and livestock production are competing/conflicting land uses, or that bee keeping is an intervention that in not preferred by households with larger herd sizes. A cross-tabulation of the intervention and occupation of respondents revealed that only 6.6%

of livestock keepers carried out bee keeping.

### *4.4.4 Extraction of Gums and Resins.*

Harvesting of gums and resins was prevalent among 8.5% of the population, 6.9% in the treatment area and 10.1% in the control area. We found a minor positive association between extraction of gums and resins with both climate resilience and household incomes. In the contrary, livestock units were negatively associated with this intervention. The households engaged in this intervention had livestock numbers less by about nine units.

### *4.4.5 Grass Seed Banking.*

A small proportion of the community practised grass seed banking, 5.9% in the entire target area, 6.6% and 5.2% in the treatment and comparison areas respectively. The intervention is positively associated with climate resilience ($p>0.050$ and household incomes ($p<0.05$). Households practising seed banking have their incomes higher by US$70.87. Regarding the effect of seed banking on livestock units, we found a negative relationship whereby households practising the intervention have their livestock less by approximately five units. This negative association may be an indication that seed banking serves as an income generating enterprise as opposed to complementing the provision of livestock fodder at the household level.

### *4.4.6 Soil and Water Conservation.*

Overall, 37.1% of the population carried out soil and water conservation, 38.7% in the treatment area and 35.4% in the control area. Households practising the interventions have higher climate resilience ($p>0.1$), higher incomes ($p>0.1$) and less livestock units ($p>0.1$). Further analysis revealed that fewer livestock farmers (16%) practise soils and water conservation compared to crop farmers, 58% of whom undertake the intervention. This indicates that soil and water conservation was mainly for improving crop production.

### *4.4.7 Value Addition of Crop and Livestock Products.*

Approximately one third of the population practice value addition, 29.2% in the whole study area, 25.5% in the treatment areas and 33.0% in control area. The intervention is positively associated with climate resilience ($p<0.001$), incomes ($p=0.001$) and lower livestock units ($p>0.1$). The main occupations of households practising value addition were crop farmers (33.5%), livestock keepers (30.3%) and small-scale businesses/enterprises (27.5%).

### *4.4.8 Sustainable Nature-based Enterprises* (e.g. tourism/eco-tourism)

Generally, tourism activities were very low, representing less than 5.0% in both the treatment and comparison areas. Despite this, the intervention was associated with increased climate resilience ($p<0.05$) and higher incomes ($p<0.05$). For instance, households involved in tourism activities have incomes higher by US$38.52. We however found that households engaging in tourism activities have less livestock units ($p=0.007$). This negative trend seems to indicate that livestock keepers with larger herd sizes do not prefer to engage in tourism activities. Could this be a pointer to incompatibility or conflicting land uses between livestock production and toursim?

### *4.4.9 Training in Climate Change, Sustainable Land management or Natural Resource Management.*

The proportion of households that had received this training was low, being 13.9% in the entire project area, 11.6%, and 16.2% respectively in the treatment and control areas. Households that had received training in climate change, sustainable land management or natural resource management had enhanced climate resilience ($p<0.001$), incomes ($p>0.5$) and livestock units ($p=0.41$). Training activities were associated with increased income of US$29.05 and livestock numbers of approximately 14 units.

## 5.0 Significance and Implications of Results

This study has successfully identified a valid control group for comparison with the TWENDE project treatment group in the assessment of project impact. In addition to defining the comparison group, the study has also established the baseline benchmarks for the expected project outcome indicators by providing an estimate of first difference in outcomes between the two study groups. The findings of this study have implications on planning and implementation of the TWENDE project. Interventions that will bring about causality in the three expected outcomes have been identified and therfore, planning and implementation among the the treatment group needs to be well aligned with these interventions. For instance, the baseline study has shown the different groups that prefer to enroll in the respective project interventions hence the projet should specifically target such groups during project implementation. It has also been shown that the treatment area has lower implementation or uptake rate for most of the planned interventions thus justifying the selection of the project beneficiaries. There has been proof of the validity of design as the study has shown that the interventions planned for implementation have positive relationship with the project outcomes.

## 6.0 Conclusions

We have drawn four main conclusions from the study:

(i) Based on the positive balancing property of propensity scores test results, it is concluded that the control group identified by the study is a valid comparison group with characteristics that match those of the treatment group;

(ii) At pre-project intervention, households in the control group have higher livestock-based assets, incomes and climate resilience compared to the TWENDE project treatment group. This implies that the project is intervening in the more vulnerable communities;

(iii) The design of project intervention is still valid as most of the planned project interventions have been shown to have a positive contribution to project outcomes;

(iv) Household's main occupation influences the choice of project interventions in which to enrol or participate.

**Acknowledgement:**

The baseline study was the result of the combined efforts and invaluable contributions of TWENDE project partners and stakeholders including AFREDEC Consulting who conducted the baseline study and produced the baseline study report; IUCN TWENDE project team that provided the overall coordination of the study; the local communities and representatives of governmental and non-governmental organizations who were the research respondents (sources of information); and Green Climate Fund (GCF), the financiers of TWENDE project through the IUCN and the Government of Kenya.

**Disclaimer:**

The opinions, interpretations or conclusions expressed in this publication do not in any way reflect the opinions, views or positions of International Union for Conservation of Nature (IUCN), The Government of Kenya (GoK), the Green Climate Fund (GCF) or TWENDE project implementing partners, staff or their affiliates. The publication is solely based on the author's analysis, synthesis and interpretation of the TWENDE project baseline study data and report (IUCN, A Baseline Study Report of the Towards Ending Drought Emergencies (TWENDE) Project., 2021).

## References


Asmamaw, M., Mereta, S., & Ambelu, A. (2019). Exploring households' resilience to climate change-induced shocks using Climate Resilience Index in Dinki watershed, central highlands of Ethiopia. *PLoS ONE*, 14(7): e0219393.

Dassanayake, W., Mohapatra, S., K, L. M., & Adamowicz, W. (2017). Households' responses to climate change: Contingent behavior evidence from rural South Africa. *Environment and Development Economics*, 1-26.

Faul, F., & Edgar, E. (2007). G*Power 3: A flexible statistical power analysis program for the social, behavioral, and Biomedical Sciences. *Behavior Research Methods, 39 (2)*, 175-191.

Gertler, P., Martinez, S., Premand, P., L.B., R., & Vermeersc, C. (2016). *Impact Evaluation in Practice.* (Second Edition ed.). Washington, DC: Inter-American Development Bank and World Bank. doi:10.1596/978-1-4648-0779-4. License: Creative Commons Attribution CC BY 3.0 IGO.

Gori Maia, A., Burney, J., Morales Martı´nez, J., & Cesano, D. (2021). Improving production and quality of life for smallholder farmers through a climate resilience program: An experience in the Brazilian Sertão. *LoS ONE, 16(5): e0251531.P.*

IPCC. (2021). *Summary for Policymakers. In: Climate Change 2021: The Physical Science Basis. Contribution of Working Group I to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change.*

IUCN. (2015). *TWENDE Towards Ending Drought Emergencies: Ecosystem Based Adaptation in Kenya's Arid and Semi-Arid Rangelands.*

IUCN. (2021). *A Baseline Study Report of the Towards Ending Drought Emergencies (TWENDE) Project.*

J.M, K. (n.d.). Impact of Post-harvest Loss Interventions on Post-Harvest Losses of Maize among Small Holder Farmers in Tanzania: A Difference in Difference (DID) Analysis. *OIDA International Journal of Sustainable Development.* .

Kiani, A., Sardar, A., Khan, W., He, Y., Bilgic, A., Kuslu, Y., & Raja, M. (2021). Role of Agricultural Diversification in Improving Resilience to Climate Change: An Empirical Analysis with Gaussian Paradigm. . *Sustainability 2021, 13, 9539. https://doi.org/10.3390/su13179539.*

Kiiru, J. (2019). Impact of Post-harvest Loss Interventions on Post-Harvest Losses of Maize among Small Holder Farmers in Tanzania: A Difference in Difference (DID) Analysis. *OIDA International Journal of Sustainable Development, Ontario International Development Agency, Canada.*(12:12).

McPeak, J., & Little, P. (2017). Applying the concept of resilience to pastoralist household data. *Pastoralism: Research, Policy and Practice., 7:14.*

Mu, R., & Van de Walle, D. (2011). Rural Roads and Local Market Development in Vietnam. *Journal of Development Studies , 47 (5):* , 709–34.

Schwartz, H., Shaabani, S., & and Walther, D. (1991). *Range management handbook of Kenya.* . Nairobi: Ministry of Livestock Development.

SEI. (2009). *The Economics of Climate Change in East Africa. Bouma, M. Kovats, S and Hunt, A;. In. Watkiss, P., et al. Report for DFID and DANIDA. Available at http://kenya.cceconomics.org/.* .

Spratt, D., & Dunlop, I. (2019). *Existential Cimate-related Security Risk: A scenario approach.* Melbourne, Australia: Breakthrough - National Centre for Climate Restoration.

Watkiss, P., Downing, T., & Dyszynski, J. (2010). *AdaptCost Project: Analysis of.* Retrieved from https://assets.publishing.service.gov.uk/media/57a08b1240f0b652dd000aa2/60871-SEI.pdf

White, H., & Raitzer, D. (2017). *Impact Evaluation of development Interventions. A Practical Guide.* 6 ADB Avenue, Mandaluyong City, 1550 Metro Manila, Philippines: Asian Development Bank.

White, H., & Raitzer, D. A. (2017). *Impact evaluation of development interventions- A practical guide.*